MATH203 – APPLIED LINEAR ALGEBRA
HOMEWORK 4

Your Name: ______________________________________

Your Student Number: ________________________________

Your Tutorial TA: ___________________________________

Instructions.

(1) Show your work and justify your steps on every question.
(2) If a True/False statement is false, give a counterexample.
(3) You may use course materials, but your submitted solution must be written independently.
(4) Whenever a final answer is a concrete matrix, write the matrix entry-wise. A final answer left only as a product of matrices receives no credit, unless the question explicitly asks for a factorization such as a diagonalization, SVD, or MM^T form.

1. (20 points)

Answer True or False. If False, give a counterexample.

1.1) If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable, meaning it can be written as $A = PDP^{-1}$ for some diagonal matrix D.

1.2) If A is positive definite, meaning $x^T Ax > 0$ for every nonzero real vector x, then every diagonal entry a_{ii} is positive.

1.3) If A is Hermitian, meaning $A = A^H$, then the singular values of A, meaning the square roots of the eigenvalues of $A^H A$, are the absolute values of the eigenvalues of A.

1.4) If A is diagonalizable over $\mathbb{R}$, meaning $A = PDP^{-1}$ with real matrices P, D, and all eigenvalues are real, then A is symmetric, meaning $A = A^T$.

1.5) If A and B are both positive definite, meaning $x^T Ax > 0$ and $x^T Bx > 0$ for every nonzero real vector x, then $A + B$ is positive definite.

1.6) If A is real and skew-symmetric, meaning all entries of A are real and $A^T = -A$, then $I + A$ is invertible, meaning $(I + A)^{-1}$ exists.

1.7) If X is skew-Hermitian, meaning $X^H = -X$, then

$$U = (I + X)(I - X)^{-1}$$

is unitary, meaning $U^H U = I$.

1.8) If A is normal, meaning $AA^H = A^H A$, then eigenvectors corresponding to distinct eigenvalues are orthogonal, meaning their Hermitian inner product is zero.

1.9) If A is diagonalizable, meaning $A = PDP^{-1}$ for some diagonal matrix D, and all eigenvalues of A are positive real numbers, then A is positive definite, meaning $x^T Ax > 0$ for every nonzero real vector x.

1.10) If A is real, meaning all entries of A are real, and has a non-real eigenvalue $a + bi$ with $b \neq 0$, then $a - bi$ is also an eigenvalue of A.

2. (20 points)

For each matrix, compute both the projection form and the diagonalization form.

Matrix A. Let

$$A = \begin{pmatrix} -1 & 6 \\ -3 & 8 \end{pmatrix}.$$

(a) Find the characteristic polynomial $\det(tI - A)$.

(b) Find the spectral decomposition

$$A = \lambda_1 P_1 + \lambda_2 P_2,$$

where P_1 and P_2 are spectral projections. They are not required to be orthogonal projections.

(c) Find a diagonalization $A = PDP^{-1}$. Explicitly write P, D, and P^{-1}.

(d) Use either the spectral decomposition or diagonalization to compute A^{10}. Your final answer must be written as an entry-wise matrix. Leaving the answer as a product of matrices receives no credit.

Matrix B. Let

$$B = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

(e) Find the characteristic polynomial $\det(tI - B)$.

(f) Find the spectral decomposition of B

$$B = \lambda_1 P_1 + \lambda_2 P_2,$$

where P_1 and P_2 are orthogonal projections.

(g) Diagonalize B by an orthogonal matrix:

$$B = QDQ^T.$$

Explicitly write Q, D, and Q^T.

3. (15 points)

Consider the matrix

$$A = \begin{pmatrix} 2 & * & 4 \\ * & 5 & * \\ -1 & -2 & * \end{pmatrix},$$

where the four entries marked $*$ are unknown. Suppose A is diagonalizable and

$$\det(tI - A) = t^2(t - \lambda).$$

(a) Find λ.

(b) Complete the matrix by determining all entries marked $*$.

(c) Find the spectral decomposition of A.

(d) Find a diagonalization $A = PDP^{-1}$. Explicitly write P, D, and P^{-1}.

4. (20 points)

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(a) Find the characteristic polynomial $\det(tI - A^T A)$.

(b) Find the singular values of A.

(c) Find a singular value decomposition $A = U\Sigma V^T$.

5. (20 points)

Solve the system

$$\mathbf{x}' = A\mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \qquad A = \begin{pmatrix} 7 & -6 & 10 \\ 19 & -13 & 20 \\ 6 & -6 & 11 \end{pmatrix}.$$

This matrix has one integer eigenvalue and one complex-conjugate pair of eigenvalues with integer real and imaginary parts. It is not triangular or block triangular, so the problem cannot be solved by simply reading off a visible block.

(a) Find the characteristic polynomial $\det(tI - A)$.

(b) Find all eigenvalues of A.

(c) Find the spectral decomposition of A over $\mathbb{C}$.

(d) Use the spectral decomposition to write a formula for e^{At} in real form using exponentials, sine, and cosine. Then solve the initial value problem by computing

$$\mathbf{x}(t) = e^{At} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

The final answers for both e^{At} and $\mathbf{x}(t)$ must be written entry-wise. Leaving e^{At} only as a product or sum of matrices receives no credit.

(e) Describe the behavior of $\mathbf{x}(t)$ as $t \to \infty$. Your answer should distinguish the real eigenvalue mode from the oscillatory complex-eigenvalue modes.

6. (10 points)

(a) Determine whether

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

is positive definite, positive semidefinite but not positive definite, or neither. If it is positive semidefinite, write

$$B = MM^T$$

for some real matrix M.

(b) Prove: if A is an $n \times n$ positive definite matrix and C is an $m \times n$ matrix with $\operatorname{rank}(C) = n$, then $C^T AC$ is positive definite.

(c) Let

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$$

from Problem 2. Find a positive definite matrix B such that $B^2 = A$.

7. (10 points)

Let A be a real $n \times n$ matrix, diagonalizable over $\mathbb{C}$. In the spectral decomposition

$$A = \sum \lambda_j P_j,$$

suppose $\lambda = a + bi$ with $b \neq 0$ is a complex eigenvalue with eigenprojection $P = X + Yi$, where X, Y are real $n \times n$ matrices. Since A is real, $\bar{\lambda} = a - bi$ is also an eigenvalue with eigenprojection $\bar{P} = X - Yi$.

You may use the following properties without proof:

$$P^2 = P, \qquad P\bar{P} = 0.$$

Prove the following:

(a) $XY = YX$.

(b) $2X$ is a projection; that is, $(2X)^2 = 2X$.

(c) $Y^2 = -\frac{1}{2}X$.

(d) $2XY = Y$.

(e) $\mathrm{Col}(Y) \subseteq \mathrm{Col}(X)$.